

CAMPINAS, Brazil

WMO: 837210

Lat: 23.007S

Lon: 47.135W

Elev: 2170

StdP: 13.58

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.2	51.4	37.3	35.4	71.2	40.8	40.7	70.6	27.5	60.7	24.3	61.8	10.3	120	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.7	92.8	68.4	91.0	68.9	89.3	69.1	74.8	83.4	73.9	82.7	73.2	82.0	8.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	133.5	76.6	71.8	127.4	75.9	71.3	125.1	75.6	39.9	84.4	39.1	82.7	38.4	82.2	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	22.5	20.5	DB	43.3	96.4	4.0	2.4	40.5	98.1	38.1	99.5	35.9	100.8	33.0	102.5	(4)
(5)			WB	41.7	76.8	4.0	1.6	38.8	78.0	36.5	78.9	34.2	79.8	31.3	81.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.8	76.1	76.8	75.5	72.8	67.4	65.7	65.6	68.1	70.7	73.3	73.7	75.8		
	DBStd	5.80	3.64	3.35	3.14	3.85	4.30	4.32	4.71	4.92	5.29	4.92	4.04	3.32		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	166	0	0	0	2	26	41	51	29	13	3	1	0		
	CDD50	7944	809	750	792	685	538	472	483	561	621	722	711	800		
	CDD65	2632	344	330	327	237	99	63	68	124	184	259	262	335		
	CDH74	19511	2369	2412	2234	1603	598	423	567	1173	1647	2155	1914	2416		
	CDH80	6675	878	935	770	464	86	33	73	324	653	898	671	890		
(14) Wind	WSAvg	9.6	8.7	8.4	9.0	9.2	8.9	8.6	9.3	10.2	11.2	11.1	10.8	9.4		
(15) Precipitation	PrecAvg	48.6	9.1	7.3	5.7	1.9	2.1	1.4	1.3	1.0	2.5	3.9	5.0	6.7		
	PrecMax	60.6	13.9	13.8	12.7	4.7	4.4	7.2	5.2	2.3	7.0	6.9	7.8	10.7		
	PrecMin	33.9	4.9	3.2	2.0	0.0	0.1	0.0	0.0	0.0	0.0	1.1	1.9	1.6		
	PrecStd	6.0	2.8	2.4	2.6	1.2	1.1	1.8	1.3	0.8	2.0	1.7	1.9	2.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	93.4	91.4	89.5	84.6	82.7	84.4	89.5	93.5	95.4	91.8	92.9		
		MCWB	70.7	68.7	71.7	69.3	66.8	63.9	62.7	63.3	65.3	67.4	68.9	69.0		
	2%	DB	90.0	91.0	89.2	87.4	82.5	80.7	82.3	86.2	89.9	91.8	89.4	89.9		
		MCWB	70.8	70.4	71.5	69.3	66.2	63.6	62.2	62.5	64.1	67.5	69.3	69.8		
	5%	DB	87.9	88.2	86.4	84.6	80.3	78.8	80.2	84.0	87.5	89.2	86.4	87.8		
		MCWB	70.8	70.7	70.9	68.8	65.2	63.3	61.8	62.3	64.1	67.5	68.8	69.9		
	10%	DB	85.7	86.1	84.5	82.6	77.3	76.7	77.2	80.8	84.1	85.7	84.2	85.6		
		MCWB	71.3	71.2	70.8	68.3	64.3	62.5	61.1	61.5	64.1	67.6	68.9	70.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	75.9	75.3	73.8	70.7	68.3	68.3	68.6	70.9	73.6	75.3	75.1		
		MCDB	84.0	85.5	84.9	81.8	78.5	76.2	74.6	78.8	80.4	82.1	81.4	84.8		
	2%	WB	74.4	74.6	73.9	72.4	68.7	66.5	66.3	69.0	71.8	73.4	73.8			
		MCDB	82.9	83.6	83.2	81.0	77.0	74.5	74.4	76.7	79.2	82.9	81.5	82.8		
	5%	WB	73.5	73.6	72.9	71.2	67.3	65.2	64.1	64.7	67.7	70.6	71.9	72.8		
		MCDB	82.1	82.6	82.0	79.6	75.6	73.6	74.4	76.0	78.2	81.1	80.5	81.5		
	10%	WB	72.6	72.7	71.9	70.3	66.0	63.8	62.8	63.4	66.3	69.3	70.8	71.8		
		MCDB	80.8	81.2	80.8	78.1	73.5	72.5	72.4	74.9	76.2	78.9	79.0	80.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	17.7	17.6	18.4	18.1	19.3	20.5	22.3	21.1	20.1	18.9	18.2		
		MCDBR	21.1	21.3	20.8	21.1	21.4	22.4	24.7	26.8	27.3	26.5	23.2	22.4		
	5% WB	MCWBR	6.8	6.7	6.8	6.7	7.4	8.5	8.9	8.6	8.1	7.5	7.9	7.4		
		MCWBR	17.0	17.9	18.1	17.8	18.3	18.6	20.1	22.1	20.9	20.4	18.1	17.7		
(40) Clear-Sky Solar Irradiance	taub		0.403	0.404	0.386	0.365	0.346	0.351	0.364	0.399	0.486	0.448	0.428	0.411		
		taud	2.420	2.397	2.452	2.482	2.509	2.483	2.427	2.301	2.048	2.224	2.293	2.369		
	Ebn at Noon		299	294	291	284	276	268	267	268	257	278	290	297		
		Edh at Noon	40	40	36	32	29	29	31	38	53	47	45	42		
(44) All-Sky Solar Radiation	RadAvg		1716	1803	1638	1492	1210	1124	1187	1433	1560	1721	1791	1840		
	RadStd		240	154	144	110	84	101	108	108	164	161	105	125		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-0.45	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.54	-0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page